

Geometry Dash Game (Scratch)

Introduction

This project is a Scratch-based recreation of the popular rhythm platformer **Geometry Dash**. The game focuses on precise timing, obstacle navigation, and smooth continuous movement synced with gameplay mechanics. The objective is to control a player block and avoid obstacles by jumping at the correct time to reach the end of the level.

This project was developed as part of a learning exercise to understand **game logic, event-driven programming, and collision detection** using Scratch.

Objective

- To design a playable Geometry Dash-style game using Scratch
- To implement smooth horizontal movement and gravity-based jumping
- To learn game mechanics such as collision detection, scoring, and level reset
- To understand how complex games can be built using visual block programming

Programming Logic (Scratch Blocks)

Player Movement Logic

- Continuous movement using change x by blocks
- Jump action triggered using keyboard input (space key)
- Gravity simulated using repeated downward motion when not jumping

Code Reference:

Screenshot of Scratch blocks showing continuous movement and gravity logic (adapted from Griffpatch's approach).

Jump Mechanism

- Jump occurs only when the player is on the ground
- A temporary upward velocity is applied
- Gravity gradually pulls the player back down

Code Reference:

Screenshot of Scratch blocks handling jump velocity and ground detection.

Collision Detection

- Obstacles are detected using touching color or touching sprite blocks
- When a collision occurs:
 - The game resets
 - Player returns to the starting position

Code Reference:

Screenshot of Scratch blocks checking collision and resetting the game.

Level Reset Logic

- Player position is reset
- Level elements return to initial state
- Score or progress is reset

Code Reference:

Screenshot of Scratch blocks used for resetting the game state.

User Controls

- **Space key** – Jump
- The player moves automatically; no directional input is required

Key Learning Outcomes

- Understanding of **event-based programming**
- Implementation of **game physics** using Scratch blocks
- Use of **loops, conditions, and variables**
- Learning how complex gameplay can be created without traditional coding languages

Conclusion

This Geometry Dash Scratch project demonstrates how advanced game mechanics can be achieved using visual programming. By learning from Griffpatch's tutorials and adapting the logic independently, this project helped strengthen problem-solving skills and understanding of game development fundamentals.

The project serves as a strong foundation for transitioning into more advanced game development or programming environments.

